

Excel VBA Expenses Dashboard v2 (FY 2025)

Complete VBA module with live progress tracking, status bar animation, and headings-on layout.

Workbook Prerequisites

Before running the macro, your workbook needs a sheet named **Data** with these exact headers in Row 1 (order does not matter): Expense ID, Date, Department, Category, Amount (INR), Payment Method, Approved, Budget Allocated, Budget Remaining. A **Month** column is auto-added by the macro if missing.

How to Use

1. Open your Excel workbook and ensure the Data sheet exists with the required headers.
2. Press Alt+F11 to open the VBA editor.
3. Insert a new standard module (Insert > Module).
4. Paste the complete code (below) into the module.
5. Close the VBA editor.
6. Press Alt+F8, select BuildExpensesDashboard, click Run.
7. Watch the status bar for live progress. A success message confirms completion.

Complete VBA Module

Option Explicit

```
Private Const DATA_SHEET As String = "Data"
Private Const PIVOT_SHEET As String = "Pivot"
Private Const DASH_SHEET As String = "Dashboard"
Private Const TABLE_NAME As String = "tblExpenses"
Private Const MONEY_K_FORMAT As String = "#,##0,""K"""
```

Private BuildStep As String

```
'=====
' SECTION 1 - Progress Helpers
'=====
```

```
Private Sub ShowProgress(ByVal msg As String, _
                        Optional ByVal flashSheet As String = "", _
                        Optional ByVal pauseMs As Long = 350)
    Application.StatusBar = " [*] " & msg
    DoEvents
    If flashSheet <> "" Then
        On Error Resume Next
        ThisWorkbook.Worksheets(flashSheet).Activate
        DoEvents
        On Error GoTo 0
    End If
    If pauseMs > 0 Then Wait pauseMs
End Sub
```

```
Private Sub Wait(ByVal ms As Long)
    Dim t As Single
    t = Timer
    Do While (Timer - t) * 1000 < ms
        DoEvents
    Loop
End Sub
```

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'=====
' SECTION 2 - Public Entry Point
'=====
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```
Public Sub BuildExpensesDashboard()
    On Error GoTo CleanFail

    BuildStep = "Starting"
    Application.ScreenUpdating = True
    Application.DisplayAlerts = False
    Application.EnableEvents = False
    Application.Calculation = xlCalculationAutomatic

    ShowProgress "Opening Data sheet...", DATA_SHEET, 300

    BuildStep = "Preparing Data"
    ShowProgress "Step 1 of 3 - Formatting Data sheet...", DATA_SHEET, 200
    PrepareDataSheet

    BuildStep = "Building Pivots"
    ShowProgress "Step 2 of 3 - Building PivotTables...", "", 200
    BuildPivotSheet

    BuildStep = "Building Dashboard"
    ShowProgress "Step 3 of 3 - Building Dashboard...", "", 200
    BuildDashboardSheet

    BuildStep = "Final Refresh"
    ShowProgress "Refreshing all data...", DASH_SHEET, 400
    ThisWorkbook.RefreshAll
    Application.CalculateFull
    Application.Calculation = xlCalculationAutomatic

```

```
CleanExit:
    Application.StatusBar = False
    Application.EnableEvents = True
    Application.DisplayAlerts = True
    Application.ScreenUpdating = True
    MsgBox "Dashboard Created Successfully", vbInformation, "Done"
    Exit Sub

```

```
CleanFail:
    Application.StatusBar = False
    Application.Calculation = xlCalculationAutomatic

```

```

Application.EnableEvents = True
Application.DisplayAlerts = True
Application.ScreenUpdating = True
MsgBox "Dashboard build failed at step: " & BuildStep & vbCrLf & _
    "Error " & Err.Number & ": " & Err.Description, vbCritical
End Sub

'=====
' SECTION 3 - PrepareDataSheet
'=====
Private Sub PrepareDataSheet()
    Dim ws As Worksheet, lo As ListObject, rng As Range
    Dim lastRow As Long, lastCol As Long

    Set ws = ThisWorkbook.Worksheets(DATA_SHEET)
    ws.Activate

    ShowProgress "Checking required columns...", , 250
    RequireHeader ws, "Expense ID"
    RequireHeader ws, "Date"
    RequireHeader ws, "Department"
    RequireHeader ws, "Category"
    RequireHeader ws, "Amount (INR)"
    RequireHeader ws, "Payment Method"
    RequireHeader ws, "Approved"
    RequireHeader ws, "Budget Allocated"
    RequireHeader ws, "Budget Remaining"

    ShowProgress "Inserting Month column...", , 300
    If HeaderColumn(ws, "Month") = 0 Then
        ws.Cells(1, LastUsedCol(ws) + 1).Value = "Month"
    End If

    ShowProgress "Creating Excel Table (tblExpenses)...", , 350
    lastRow = LastUsedRow(ws)
    lastCol = LastUsedCol(ws)
    Set rng = ws.Range(ws.Cells(1, 1), ws.Cells(lastRow, lastCol))

    Dim lo2 As ListObject
    On Error Resume Next
    Set lo2 = ws.ListObjects(TABLE_NAME)
    On Error GoTo 0
    If lo2 Is Nothing Then
        Set lo = ws.ListObjects.Add(xlSrcRange, rng, , xlYes)
        lo.Name = TABLE_NAME
    Else
        Set lo = lo2
        lo.Resize rng
    End If

    lo.TableStyle = "TableStyleMedium4"
    lo.ShowAutoFilterDropDown = True
    DoEvents

    ShowProgress "Writing Month formula in every row...", , 300
    lo.ListColumns("Month").DataBodyRange.Formula = "=TEXT([@Date],""mmm"")"
    DoEvents

    ShowProgress "Formatting Date column...", , 250
    SafeNumberFormat lo.ListColumns("Date").DataBodyRange, "dd-mmm-yyyy"
    DoEvents

    ShowProgress "Formatting Amount (INR) column...", , 250
    SafeNumberFormat lo.ListColumns("Amount (INR)").DataBodyRange, MONEY_K_FORMAT
    DoEvents

    ShowProgress "Formatting Budget Allocated column...", , 250
    SafeNumberFormat lo.ListColumns("Budget Allocated").DataBodyRange, MONEY_K_FORMAT
    DoEvents

    ShowProgress "Formatting Budget Remaining column...", , 250
    SafeNumberFormat lo.ListColumns("Budget Remaining").DataBodyRange, MONEY_K_FORMAT
    DoEvents

    ShowProgress "Styling header row...", , 300
    With ws.Rows(1)
        .Font.Bold = True
        .Interior.Color = RGB(15, 76, 31)
        .Font.Color = RGB(255, 255, 255)
    End With

```

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DoEvents

ShowProgress "Auto-fitting column widths...", , 250
ws.Columns.AutoFit
DoEvents
End Sub

'=====
' SECTION 4 - BuildPivotSheet
'=====
Private Sub BuildPivotSheet()
    Dim wsp As Worksheet, lo As ListObject, pc As PivotCache

    ShowProgress "Removing old sheets...", , 300
    DeleteSheetIfExists DASH_SHEET
    DeleteSheetIfExists PIVOT_SHEET

    ShowProgress "Creating Pivot sheet...", , 300
    Set wsp = ThisWorkbook.Worksheets.Add(After:=ThisWorkbook.Worksheets(DATA_SHEET))
    wsp.Name = PIVOT_SHEET
    wsp.Activate
    DoEvents

    Set lo = ThisWorkbook.Worksheets(DATA_SHEET).ListObjects(TABLE_NAME)

    Set pc = ThisWorkbook.PivotCaches.Create( _
        SourceType:=xlDatabase, _
        SourceData:=lo.Range.Address(True, True, xlR1C1, True))

    ShowProgress "Building PivotTable 1/7 - Expense by Department...", PIVOT_SHEET, 400
    CreatePivot_ExpenseByDepartment pc, wsp.Range("A3")
    DoEvents

    ShowProgress "Building PivotTable 2/7 - Expense by Category...", PIVOT_SHEET, 400
    CreatePivot_ExpenseByCategory pc, wsp.Range("D3")
    DoEvents

    ShowProgress "Building PivotTable 3/7 - Monthly Trend...", PIVOT_SHEET, 400
    CreatePivot_MonthlyTrend pc, wsp.Range("G3")
    DoEvents

    ShowProgress "Building PivotTable 4/7 - Approved Status...", PIVOT_SHEET, 400
    CreatePivot_ApprovedStatus pc, wsp.Range("J3")
    DoEvents

    ShowProgress "Building PivotTable 5/7 - KPI Totals...", PIVOT_SHEET, 400
    CreatePivot_KPI pc, wsp.Range("M3")
    DoEvents

    ShowProgress "Building PivotTable 6/7 - Payment Method...", PIVOT_SHEET, 400
    CreatePivot_PaymentMethod pc, wsp.Range("A25")
    DoEvents

    ShowProgress "Building PivotTable 7/7 - Budget vs Actual...", PIVOT_SHEET, 400
    CreatePivot_BudgetVsActual pc, wsp.Range("D25")
    DoEvents

    ShowProgress "Generating chart-helper data ranges...", PIVOT_SHEET, 350
    BuildChartHelpers wsp
    DoEvents

    wsp.Columns.AutoFit
End Sub

'=====
' SECTION 5 - Seven PivotTable Builders
'=====
Private Sub CreatePivot_ExpenseByDepartment(ByVal pc As PivotCache, ByVal dest As Range)
    Dim pt As PivotTable
    Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_ExpenseByDepartment")
    With pt
        .PivotFields("Department").Orientation = xlRowField
        .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
        .RowAxisLayout xlTabularRow
        .RefreshTable
    End With
    FormatPivotValues pt, MONEY_K_FORMAT
End Sub

Private Sub CreatePivot_ExpenseByCategory(ByVal pc As PivotCache, ByVal dest As Range)

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Dim pt As PivotTable
Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_ExpenseByCategory")
With pt
    .PivotFields("Category").Orientation = xlRowField
    .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
    .RowAxisLayout xlTabularRow
    .RefreshTable
End With
FormatPivotValues pt, MONEY_K_FORMAT
End Sub

Private Sub CreatePivot_MonthlyTrend(ByVal pc As PivotCache, ByVal dest As Range)
Dim pt As PivotTable
Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_MonthlyTrend")
With pt
    .PivotFields("Month").Orientation = xlRowField
    .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
    .PivotFields("Month").AutoSort xlAscending, "Month"
    .RowAxisLayout xlTabularRow
    .RefreshTable
End With
FormatPivotValues pt, MONEY_K_FORMAT
End Sub

Private Sub CreatePivot_ApprovedStatus(ByVal pc As PivotCache, ByVal dest As Range)
Dim pt As PivotTable
Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_ApprovedStatus")
With pt
    .PivotFields("Approved").Orientation = xlRowField
    .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
    .AddDataField .PivotFields("Expense ID"), "Count of Expense ID", xlCount
    .RowAxisLayout xlTabularRow
    .RefreshTable
End With
On Error Resume Next
pt.DataFields(1).DataRange.NumberFormat = MONEY_K_FORMAT
pt.DataFields(2).DataRange.NumberFormat = "#,##0"
On Error GoTo 0
End Sub

Private Sub CreatePivot_KPI(ByVal pc As PivotCache, ByVal dest As Range)
Dim pt As PivotTable
Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_KPI")
With pt
    .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
    .AddDataField .PivotFields("Budget Allocated"), "Sum of Budget Allocated", xlSum
    .AddDataField .PivotFields("Budget Remaining"), "Sum of Budget Remaining", xlSum
    .AddDataField .PivotFields("Expense ID"), "Count of Expense ID", xlCount
    .RefreshTable
End With
FormatPivotValues pt, MONEY_K_FORMAT
End Sub

Private Sub CreatePivot_PaymentMethod(ByVal pc As PivotCache, ByVal dest As Range)
Dim pt As PivotTable
Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_PaymentMethod")
With pt
    .PivotFields("Payment Method").Orientation = xlRowField
    .AddDataField .PivotFields("Amount (INR)", "Sum of Amount (INR)", xlSum
    .RowAxisLayout xlTabularRow
    .RefreshTable
End With
FormatPivotValues pt, MONEY_K_FORMAT
End Sub

Private Sub CreatePivot_BudgetVsActual(ByVal pc As PivotCache, ByVal dest As Range)
Dim pt As PivotTable
Set pt = pc.CreatePivotTable(TableDestination:=dest, TableName:="PT_BudgetVsActual")
With pt
    .PivotFields("Department").Orientation = xlRowField
    .AddDataField .PivotFields("Amount (INR)", "Actual Spent", xlSum
    .AddDataField .PivotFields("Budget Allocated"), "Sum of Budget Allocated", xlSum
    .RowAxisLayout xlTabularRow
    .RefreshTable
End With
FormatPivotValues pt, MONEY_K_FORMAT
End Sub

'=====
' SECTION 6 - Chart Helper Data Builders

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'=====
Private Sub BuildChartHelpers(ByVal ws As Worksheet)
    ws.Range("A50:Z95").ClearContents

    BuildTwoColumnHelper ws, "A50", ws.PivotTables("PT_ExpenseByDepartment"), "Department", "Amount"
    BuildTwoColumnHelper ws, "D50", ws.PivotTables("PT_ExpenseByCategory"), "Category", "Amount"
    BuildTwoColumnHelper ws, "G50", ws.PivotTables("PT_MonthlyTrend"), "Month", "Amount"
    BuildTwoColumnHelper ws, "J50", ws.PivotTables("PT_PaymentMethod"), "Payment Method", "Amount"
    BuildBudgetHelper ws, "M50", ws.PivotTables("PT_BudgetVsActual")
End Sub

Private Sub BuildTwoColumnHelper(ByVal ws As Worksheet, ByVal topLeft As String, _
    ByVal pt As PivotTable, ByVal h1 As String, ByVal h2 As String)
    Dim r As Range, i As Long, n As Long
    Set r = ws.Range(topLeft)
    r.Value = h1
    r.Offset(0, 1).Value = h2
    n = PivotDataRowsCount(pt)
    For i = 1 To n
        r.Offset(i, 0).Formula = "=" & pt.RowRange.Cells(i + 1, 1).Address(True, True, xlA1, True)
        r.Offset(i, 1).Formula = "=" & pt.DataBodyRange.Cells(i, 1).Address(True, True, xlA1, True)
    Next i
    SafeNumberFormat r.Offset(1, 1).Resize(n, 1), MONEY_K_FORMAT
End Sub

Private Sub BuildBudgetHelper(ByVal ws As Worksheet, ByVal topLeft As String, ByVal pt As PivotTable)
    Dim r As Range, i As Long, n As Long
    Set r = ws.Range(topLeft)
    r.Value = "Department"
    r.Offset(0, 1).Value = "Actual Spent"
    r.Offset(0, 2).Value = "Budget Allocated"
    n = PivotDataRowsCount(pt)
    For i = 1 To n
        r.Offset(i, 0).Formula = "=" & pt.RowRange.Cells(i + 1, 1).Address(True, True, xlA1, True)
        r.Offset(i, 1).Formula = "=" & pt.DataBodyRange.Cells(i, 1).Address(True, True, xlA1, True)
        r.Offset(i, 2).Formula = "=" & pt.DataBodyRange.Cells(i, 2).Address(True, True, xlA1, True)
    Next i
    SafeNumberFormat r.Offset(1, 1).Resize(n, 2), MONEY_K_FORMAT
End Sub

'=====
' SECTION 7 - BuildDashboardSheet Orchestrator
'=====
Private Sub BuildDashboardSheet()
    Dim wSD As Worksheet

    ShowProgress "Creating Dashboard sheet...", , 300
    DeleteSheetIfExists DASH_SHEET
    Set wSD = ThisWorkbook.Worksheets.Add(Before:=ThisWorkbook.Worksheets(1))
    wSD.Name = DASH_SHEET
    wSD.Activate

    ActiveWindow.Zoom = 70
    ActiveWindow.DisplayHeadings = True
    DoEvents

    ShowProgress "Applying canvas & grid settings...", DASH_SHEET, 300
    ApplyDashboardCanvas wSD
    DoEvents

    ShowProgress "Drawing header banner...", DASH_SHEET, 400
    CreateHeader wSD
    DoEvents

    ShowProgress "Writing KPI helper formulas...", DASH_SHEET, 300
    CreateKPIHelperFormulas wSD
    DoEvents

    ShowProgress "Building KPI cards (6 cards)...", DASH_SHEET, 500
    CreateKPICards wSD
    DoEvents

    ShowProgress "Inserting charts - Department, Category, Trend, Budget, Payment...", DASH_SHEET, 600
    CreateDashboardCharts wSD
    DoEvents

    ShowProgress "Adding Month slicer...", DASH_SHEET, 400
    CreateMonthSlicer wSD
    DoEvents
    ShowProgress "Adding Department slicer...", DASH_SHEET, 400

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CreateDepartmentSlicer wsD
DoEvents

ShowProgress "Applying final formatting & print settings...", DASH_SHEET, 350
FinalDashboardFormatting wsD
DoEvents

wsD.Activate
wsD.Range("A1").Select
ActiveWindow.Zoom = 70
ActiveWindow.DisplayHeadings = True
End Sub

'=====
' SECTION 8 - ApplyDashboardCanvas
'=====
Private Sub ApplyDashboardCanvas(ByVal ws As Worksheet)
    Dim i As Long
    ws.Activate
    ActiveWindow.DisplayGridlines = False
    ws.Cells.Interior.Color = RGB(246, 250, 244)
    ws.Cells.Font.Name = "Aptos"
    ws.Cells.Font.Size = 10

    For i = 1 To 30
        ws.Columns(i).ColumnWidth = 7.3
    Next i

    ws.Rows("1:3").RowHeight = 22
    ws.Rows("4:29").RowHeight = 18
End Sub

'=====
' SECTION 9 - CreateHeader
'=====
Private Sub CreateHeader(ByVal ws As Worksheet)
    Dim shp As Shape, icon As Shape, dateBox As Shape, titleBox As Shape

    Dim bannerLeft As Double: bannerLeft = ws.Range("A1").Left
    Dim bannerTop As Double: bannerTop = ws.Range("A1").Top
    Dim bannerWidth As Double: bannerWidth = ws.Range("A1:AD3").Width
    Dim bannerHeight As Double: bannerHeight = ws.Range("A1:AD3").Height

    Dim iconW As Double: iconW = 42
    Dim iconH As Double: iconH = 42
    Dim iconLeft As Double: iconLeft = bannerLeft + 12
    Dim iconTop As Double: iconTop = bannerTop + (bannerHeight - iconH) / 2

    Dim dateW As Double: dateW = 190
    Dim dateH As Double: dateH = bannerHeight - 8
    Dim dateLeft As Double: dateLeft = bannerLeft + bannerWidth - dateW - 10
    Dim dateTop As Double: dateTop = bannerTop + 4

    Dim titleLeft As Double: titleLeft = iconLeft + iconW + 8
    Dim titleWidth As Double: titleWidth = dateLeft - titleLeft - 8
    Dim titleTop As Double: titleTop = bannerTop
    Dim titleHeight As Double: titleHeight = bannerHeight

    Set shp = ws.Shapes.AddShape(msoShapeRoundedRectangle, bannerLeft, bannerTop, bannerWidth, bannerHeight)
    With shp
        .Name = "Header_Banner"
        .Fill.TwoColorGradient msoGradientHorizontal, 1
        .Fill.ForeColor.RGB = RGB(6, 66, 25)
        .Fill.BackColor.RGB = RGB(17, 112, 31)
        .Line.Visible = msoFalse
        .TextFrame.Characters.Text = ""
    End With

    Set icon = ws.Shapes.AddShape(msoShapeOval, iconLeft, iconTop, iconW, iconH)
    With icon
        .Fill.ForeColor.RGB = RGB(12, 88, 30)
        .Line.ForeColor.RGB = RGB(230, 245, 230)
        .TextFrame.Characters.Text = "$"
        .TextFrame.HorizontalAlignment = xlHAlignCenter
        .TextFrame.VerticalAlignment = xlVAlignCenter
        .TextFrame.Characters.Font.Color = RGB(255, 255, 255)
        .TextFrame.Characters.Font.Bold = True
        .TextFrame.Characters.Font.Size = 22
    End With

    Set titleBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _

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        titleLeft, titleTop, titleWidth, titleHeight)
With titleBox
    .Fill.Visible = msoFalse
    .Line.Visible = msoFalse
    .TextFrame2.WordWrap = msoFalse
    .TextFrame2.AutoSize = msoAutoSizeNone
    .TextFrame.Characters.Text = "EXPENSES DASHBOARD | FY 2025"
    .TextFrame.HorizontalAlignment = xlHAlignCenter
    .TextFrame.VerticalAlignment = xlVAlignCenter
    .TextFrame.Characters.Font.Color = RGB(255, 255, 255)
    .TextFrame.Characters.Font.Bold = True
    .TextFrame.Characters.Font.Size = 44
End With

Set dateBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
    dateLeft, dateTop, dateW, dateH)
With dateBox
    .Fill.Visible = msoFalse
    .Line.Visible = msoFalse
    .TextFrame2.WordWrap = msoTrue
    .TextFrame2.AutoSize = msoAutoSizeNone
    .TextFrame.Characters.Text = "Data as of:" & vbCrLf & Format(Date, "mmm dd, yyyy")
    .TextFrame.HorizontalAlignment = xlHAlignCenter
    .TextFrame.VerticalAlignment = xlVAlignCenter
    .TextFrame.Characters.Font.Color = RGB(255, 255, 255)
    .TextFrame.Characters.Font.Bold = True
    .TextFrame.Characters.Font.Size = 18
End With
End Sub

'=====
' SECTION 10 - KPI Helper Formulas
'=====
Private Sub CreateKPIHelperFormulas(ByVal ws As Worksheet)
    ws.Range("AF1:AG20").ClearContents

    ws.Range("AF2").Formula = "=IFERROR(GETPIVOTDATA(""Sum of Amount (INR)"",Pivot!$M$3),0)"
    ws.Range("AF3").Formula = "=IFERROR(GETPIVOTDATA(""Sum of Budget Allocated"",Pivot!$M$3),0)"
    ws.Range("AF4").Formula = "=IFERROR(GETPIVOTDATA(""Sum of Budget Remaining"",Pivot!$M$3),0)"
    ws.Range("AF5").Formula = "=IFERROR(GETPIVOTDATA(""Count of Expense ID"",Pivot!$M$3),0)"
    ws.Range("AF6").Formula = "=IFERROR(GETPIVOTDATA(""Count of Expense ID"",Pivot!$J$3, ""Approved"", ""Yes"")/AF5,0)"
    ws.Range("AF7").Formula = "=IFERROR(AF2/AF5,0)"

    SafeNumberFormat ws.Range("AF2:AF4,AF7"), MONEY_K_FORMAT
    SafeNumberFormat ws.Range("AF5"), "#,##0"
    SafeNumberFormat ws.Range("AF6"), "0.0%"
End Sub

'=====
' SECTION 11 - KPI Cards
'=====
Private Sub CreateKPICards(ByVal ws As Worksheet)
    ShowProgress " KPI card 1/6 - Total Expenses...", DASH_SHEET, 300
    AddKPICard ws, "A5:E8", "Total Expenses", "=Dashboard!$AF$2", "$", "Total Spend", RGB(21, 125, 24)
    DoEvents
    ShowProgress " KPI card 2/6 - Budget Allocated...", DASH_SHEET, 300
    AddKPICard ws, "F5:J8", "Budget Allocated", "=Dashboard!$AF$3", "%", "100% of Plan", RGB(67, 152, 32)
    DoEvents
    ShowProgress " KPI card 3/6 - Budget Remaining...", DASH_SHEET, 300
    AddKPICard ws, "K5:O8", "Budget Remaining", "=Dashboard!$AF$4", "$", "Remaining Budget", RGB(93, 171, 57)
    DoEvents
    ShowProgress " KPI card 4/6 - Transactions...", DASH_SHEET, 300
    AddKPICard ws, "P5:T8", "Transactions", "=Dashboard!$AF$5", "#", "Total Transactions", RGB(17, 121, 58)
    DoEvents
    ShowProgress " KPI card 5/6 - Approval Rate...", DASH_SHEET, 300
    AddKPICard ws, "U5:Y8", "Approval Rate", "=Dashboard!$AF$6", "0", "Approved Items", RGB(37, 137, 48)
    DoEvents
    ShowProgress " KPI card 6/6 - Avg Expense...", DASH_SHEET, 300
    AddKPICard ws, "Z5:AD8", "Avg Expense", "=Dashboard!$AF$7", "^", "Per Transaction", RGB(22, 101, 52)
    DoEvents
End Sub

Private Sub AddKPICard(ByVal ws As Worksheet, ByVal addr As String, ByVal title As String, _
    ByVal valueFormula As String, ByVal iconText As String, _
    ByVal subTitle As String, ByVal accent As Long)
    Dim box As Range, card As Shape, icon As Shape
    Dim titleBox As Shape, valueBox As Shape, subBox As Shape

    Set box = ws.Range(addr)
    Set card = ws.Shapes.AddShape(msoShapeRoundedRectangle, box.Left, box.Top, box.Width, box.Height)

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With card
.Fill.ForeColor.RGB = RGB(255, 255, 255)
.Line.ForeColor.RGB = RGB(208, 226, 203)
.Line.Weight = 0.75
On Error Resume Next
.Shadow.Visible = msoTrue
.Shadow.ForeColor.RGB = RGB(200, 210, 200)
.Shadow.Transparency = 0.7
.Shadow.OffsetX = 1
.Shadow.OffsetY = 1
On Error GoTo 0
End With

Set icon = ws.Shapes.AddShape(msoShapeOval, box.Left + 10, box.Top + 16, 38, 38)
With icon
.Fill.TwoColorGradient msoGradientDiagonalUp, 1
.Fill.ForeColor.RGB = accent
.Fill.BackColor.RGB = RGB(8, 80, 25)
.Line.Visible = msoFalse
.TextFrame.Characters.Text = iconText
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.VerticalAlignment = xlVAlignCenter
.TextFrame.Characters.Font.Color = RGB(255, 255, 255)
.TextFrame.Characters.Font.Bold = True
.TextFrame.Characters.Font.Size = 13
End With

Set titleBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
box.Left + 54, box.Top + 8, box.Width - 58, 22)
With titleBox
.Fill.Visible = msoFalse
.Line.Visible = msoFalse
.TextFrame.Characters.Text = title
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.VerticalAlignment = xlVAlignCenter
.TextFrame.Characters.Font.Bold = True
.TextFrame.Characters.Font.Size = 18
.TextFrame.Characters.Font.Color = RGB(45, 65, 45)
End With

Set valueBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
box.Left + 54, box.Top + 29, box.Width - 58, 34)
With valueBox
.Fill.Visible = msoFalse
.Line.Visible = msoFalse
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.VerticalAlignment = xlVAlignCenter
.DrawingObject.Formula = valueFormula
.TextFrame.Characters.Font.Bold = True
.TextFrame.Characters.Font.Size = 28
.TextFrame.Characters.Font.Color = RGB(5, 74, 25)
End With

Set subBox = ws.Shapes.AddTextbox(msoTextOrientationHorizontal, _
box.Left + 54, box.Top + 58, box.Width - 58, 13)
With subBox
.Fill.Visible = msoFalse
.Line.Visible = msoFalse
.TextFrame.Characters.Text = subTitle
.TextFrame.HorizontalAlignment = xlHAlignCenter
.TextFrame.Characters.Font.Size = 8
.TextFrame.Characters.Font.Color = RGB(80, 95, 80)
End With
End Sub

'=====
' SECTION 12 - Dashboard Charts
'=====
Private Sub CreateDashboardCharts(ByVal ws As Worksheet)
Dim p As Worksheet
Set p = ThisWorkbook.Worksheets(PIVOT_SHEET)

ShowProgress " Chart 1/5 - Expense by Department...", DASH_SHEET, 450
AddNormalChart ws, xlColumnClustered, "Expense by Department", p.Range("A50").CurrentRegion, "A10:L18", "dept"
DoEvents

ShowProgress " Chart 2/5 - Expense by Category...", DASH_SHEET, 450
AddNormalChart ws, xlDoughnut, "Expense by Category", p.Range("D50").CurrentRegion, "M10:V18", "category"
DoEvents
ShowProgress " Chart 3/5 - Monthly Trend...", DASH_SHEET, 450

```

```

AddNormalChart ws, xlLineMarkers, "Monthly Trend", p.Range("G50").CurrentRegion, "A19:L28", "trend"
DoEvents

ShowProgress " Chart 4/5 - Budget vs Actual...", DASH_SHEET, 450
AddNormalChart ws, xlColumnClustered, "Budget vs Actual", p.Range("M50").CurrentRegion, "M19:V28", "budget"
DoEvents

ShowProgress " Chart 5/5 - Payment Method...", DASH_SHEET, 450
AddNormalChartFixedCm ws, xlBarClustered, "Payment Method", p.Range("J50").CurrentRegion, "W19:AD28", "payment", 7.18
DoEvents
End Sub

Private Sub AddNormalChart(ByVal ws As Worksheet, ByVal chartType As XlChartType, ByVal chartTitle As String, _
    ByVal sourceRange As Range, ByVal targetAddress As String, ByVal chartKind As String)
    Dim box As Range, co As ChartObject
    Set box = ws.Range(targetAddress)
    Set co = ws.ChartObjects.Add(box.Left, box.Top, box.Width, box.Height)
    FormatChart co, chartType, chartTitle, sourceRange, chartKind
End Sub

Private Sub AddNormalChartFixedCm(ByVal ws As Worksheet, ByVal chartType As XlChartType, ByVal chartTitle As String, _
    ByVal sourceRange As Range, ByVal targetAddress As String, _
    ByVal chartKind As String, ByVal heightCm As Double)
    Dim box As Range, co As ChartObject
    Set box = ws.Range(targetAddress)
    Set co = ws.ChartObjects.Add(box.Left, box.Top, box.Width, Application.CentimetersToPoints(heightCm))
    FormatChart co, chartType, chartTitle, sourceRange, chartKind
End Sub

'=====
' SECTION 13 - FormatChart (Master Chart Formatter)
'=====
Private Sub FormatChart(ByVal co As ChartObject, ByVal chartType As XlChartType, ByVal chartTitle As String, _
    ByVal sourceRange As Range, ByVal chartKind As String)
    Dim s As Series, i As Long
    Dim colors As Variant

    colors = Array(RGB(10,93,30), RGB(132,198,55), RGB(93,177,58), RGB(22,138,150), _
        RGB(112,48,160), RGB(97,135,36), RGB(145,160,170), RGB(237,125,49))

    With co.Chart
        .ChartType = chartType
        .SetSourceData Source:=sourceRange
        .PlotVisibleOnly = False
        .HasTitle = True
        .ChartTitle.Text = chartTitle

        On Error Resume Next
        .ChartTitle.Format.TextFrame2.TextRange.Font.Size = 12
        .ChartTitle.Format.TextFrame2.TextRange.Font.Bold = msoTrue
        .ChartTitle.Format.TextFrame2.TextRange.Font.Fill.ForeColor.RGB = RGB(5, 74, 25)
        .ChartArea.Format.Fill.ForeColor.RGB = RGB(255, 255, 255)
        .PlotArea.Format.Fill.ForeColor.RGB = RGB(255, 255, 255)
        .ChartArea.Format.Line.ForeColor.RGB = RGB(208, 226, 203)
        .ChartArea.Format.Line.Weight = 0.7
        .HasLegend = True

        If chartKind = "category" Then
            .Legend.Position = xlLegendPositionRight
            .Legend.Font.Size = 12
        ElseIf chartKind = "payment" Then
            .HasLegend = False
        Else
            .Legend.Position = xlLegendPositionBottom
        End If

        If chartKind <> "category" Then
            .Axes(xlValue).TickLabels.NumberFormat = MONEY_K_FORMAT
            .Axes(xlValue).Format.Line.ForeColor.RGB = RGB(210, 210, 210)
            .Axes(xlCategory).Format.Line.ForeColor.RGB = RGB(210, 210, 210)
            .Axes(xlValue).MajorGridlines.Format.Line.ForeColor.RGB = RGB(230, 235, 230)
            .Axes(xlValue).MajorGridlines.Format.Line.Weight = 0.5
        End If

        For Each s In .SeriesCollection
            s.HasDataLabels = True
            s.DataLabels.NumberFormat = MONEY_K_FORMAT
            s.DataLabels.Font.Size = 8
            s.DataLabels.Font.Bold = True
            s.DataLabels.Font.Color = RGB(55, 55, 55)
        Next s
    End With
End Sub

```

Next s

Select Case chartKind

Case "dept"

```
For Each s In .SeriesCollection
    s.Format.Fill.TwoColorGradient msoGradientVertical, 1
    s.Format.Fill.ForeColor.RGB = RGB(10, 93, 30)
    s.Format.Fill.BackColor.RGB = RGB(39, 142, 41)
    s.Format.Line.ForeColor.RGB = RGB(6, 66, 25)
Next s
```

Case "category"

```
.ChartGroups(1).DoughnutHoleSize = 40
For Each s In .SeriesCollection
    For i = 1 To s.Points.Count
        s.Points(i).Format.Fill.ForeColor.RGB = colors((i - 1) Mod 8)
        s.Points(i).Format.Line.ForeColor.RGB = RGB(255, 255, 255)
    Next i
    s.HasDataLabels = True
    s.DataLabels.ShowValue = True
    s.DataLabels.ShowPercentage = False
    s.DataLabels.ShowCategoryName = False
    s.DataLabels.Position = xlLabelPositionBestFit
    s.DataLabels.NumberFormat = MONEY_K_FORMAT
    s.DataLabels.Font.Size = 8
    s.DataLabels.Font.Bold = True
    s.DataLabels.Font.Color = RGB(255, 255, 255)
Next s
.PlotArea.Left = .ChartArea.Width * 0.03
.PlotArea.Top = .ChartArea.Height * 0.12
.PlotArea.Width = .ChartArea.Width * 0.55
.PlotArea.Height = .ChartArea.Height * 0.78
.Legend.Left = 229.614
.Legend.Width = 206.285
.Legend.Top = .ChartArea.Height * 0.13
.Legend.Height = .ChartArea.Height * 0.75
```

Case "trend"

```
For Each s In .SeriesCollection
    s.Format.Line.ForeColor.RGB = RGB(8, 88, 28)
    s.Format.Line.Weight = 2.25
    s.MarkerStyle = xlMarkerStyleCircle
    s.MarkerSize = 6
    s.MarkerForegroundColor = RGB(8, 88, 28)
    s.MarkerBackgroundColor = RGB(39, 142, 41)
    s.DataLabels.Font.Color = RGB(5, 74, 25)
Next s
```

Case "budget"

```
If .SeriesCollection.Count >= 1 Then .SeriesCollection(1).Format.Fill.ForeColor.RGB = RGB(10, 93, 30)
If .SeriesCollection.Count >= 2 Then .SeriesCollection(2).Format.Fill.ForeColor.RGB = RGB(166, 210, 96)
```

Case "payment"

```
On Error Resume Next
.ChartGroups(1).GapWidth = 70
On Error GoTo 0
For Each s In .SeriesCollection
    s.Format.Fill.TwoColorGradient msoGradientHorizontal, 1
    s.Format.Fill.ForeColor.RGB = RGB(10, 93, 30)
    s.Format.Fill.BackColor.RGB = RGB(39, 142, 41)
    s.DataLabels.Position = xlLabelPositionOutsideEnd
Next s
```

End Select

On Error GoTo 0

End With

```
If chartKind = "payment" Then
    co.Height = co.Height * 0.8986011428
```

End If

End Sub

```
'=====
' SECTION 14 - Month Slicer
'=====
```

```
Private Sub CreateMonthSlicer(ByVal wsD As Worksheet)
    Dim sc As SlicerCache, sl As Slicer, pt As PivotTable
    Dim t As Double, l As Double, w As Double, h As Double
```

```

On Error Resume Next
ThisWorkbook.SlicerCaches("Slicer_Month").Delete
On Error GoTo 0

Set pt = ThisWorkbook.Worksheets(PIVOT_SHEET).PivotTables("PT_ExpenseByDepartment")

On Error Resume Next
Set sc = ThisWorkbook.SlicerCaches.Add2(pt, "Month", "Slicer_Month")
If sc Is Nothing Then Set sc = ThisWorkbook.SlicerCaches.Add(pt, "Month", "Slicer_Month")
On Error GoTo 0

If sc Is Nothing Then Exit Sub

AddPivotToSlicer sc, "PT_ExpenseByDepartment"
AddPivotToSlicer sc, "PT_ExpenseByCategory"
AddPivotToSlicer sc, "PT_MonthlyTrend"
AddPivotToSlicer sc, "PT_ApprovedStatus"
AddPivotToSlicer sc, "PT_KPI"
AddPivotToSlicer sc, "PT_PaymentMethod"
AddPivotToSlicer sc, "PT_BudgetVsActual"

t = wsD.Range("W10").Top
l = wsD.Range("W10").Left
w = wsD.Range("W10:AD14").Width
h = wsD.Range("W10:AD14").Height

On Error Resume Next
Set sl = sc.Slicers.Add(wsD, , "Month", "Month", t, l, w, h)
If Not sl Is Nothing Then
    sl.Top = t : sl.Left = l : sl.Width = w : sl.Height = h
    sl.NumberOfColumns = 4
    sl.Caption = "Month"
    sl.Style = "SlicerStyleLight6"
    sl.Shape.Fill.ForeColor.RGB = RGB(247, 252, 245)
    sl.Shape.Line.ForeColor.RGB = RGB(184, 215, 176)
End If
On Error GoTo 0
End Sub

'=====
' SECTION 15 - Department Slicer
'=====
Private Sub CreateDepartmentSlicer(ByVal wsD As Worksheet)
    Dim sc As SlicerCache, sl As Slicer, pt As PivotTable
    Dim t As Double, l As Double, w As Double, h As Double

    On Error Resume Next
    ThisWorkbook.SlicerCaches("Slicer_Department").Delete
    On Error GoTo 0

    Set pt = ThisWorkbook.Worksheets(PIVOT_SHEET).PivotTables("PT_ExpenseByDepartment")

    On Error Resume Next
    Set sc = ThisWorkbook.SlicerCaches.Add2(pt, "Department", "Slicer_Department")
    If sc Is Nothing Then Set sc = ThisWorkbook.SlicerCaches.Add(pt, "Department", "Slicer_Department")
    On Error GoTo 0

    If sc Is Nothing Then Exit Sub

    AddPivotToSlicer sc, "PT_ExpenseByDepartment"
    AddPivotToSlicer sc, "PT_ExpenseByCategory"
    AddPivotToSlicer sc, "PT_MonthlyTrend"
    AddPivotToSlicer sc, "PT_ApprovedStatus"
    AddPivotToSlicer sc, "PT_KPI"
    AddPivotToSlicer sc, "PT_PaymentMethod"
    AddPivotToSlicer sc, "PT_BudgetVsActual"

    t = wsD.Range("W15").Top
    l = wsD.Range("W15").Left
    w = wsD.Range("W15:AD18").Width
    h = wsD.Range("W15:AD18").Height

    On Error Resume Next
    Set sl = sc.Slicers.Add(wsD, , "Department", "Department", t, l, w, h)
    If Not sl Is Nothing Then
        sl.Top = t : sl.Left = l : sl.Width = w : sl.Height = h
        sl.NumberOfColumns = 6
        sl.Caption = "Department"
        sl.Style = "SlicerStyleLight6"
        sl.Shape.Fill.ForeColor.RGB = RGB(247, 252, 245)
    End If
End Sub

```

```

        sl.Shape.Line.ForeColor.RGB = RGB(184, 215, 176)
    End If
    On Error GoTo 0
End Sub

'=====
' SECTION 16 - FinalDashboardFormatting
'=====
Private Sub FinalDashboardFormatting(ByVal ws As Worksheet)
    ws.Range("A1:AD29").Font.Name = "Aptos"
    ws.Range("A1:AD29").VerticalAlignment = xlCenter
    ws.Columns("AF:AG").Hidden = True

    ActiveWindow.DisplayGridlines = False
    ws.PageSetup.Zoom = False
    ws.PageSetup.FitToPagesWide = 1
    ws.PageSetup.FitToPagesTall = 1
End Sub

'=====
' SECTION 17 - Shared Utility Functions
'=====
Private Sub AddPivotToSlicer(ByVal sc As SlicerCache, ByVal ptName As String)
    On Error Resume Next
    sc.PivotTables.AddPivotTable ThisWorkbook.Worksheets(PIVOT_SHEET).PivotTables(ptName)
    On Error GoTo 0
End Sub

Private Function PivotDataRowsCount(ByVal pt As PivotTable) As Long
    Dim n As Long
    On Error Resume Next
    n = pt.DataBodyRange.Rows.Count
    On Error GoTo 0
    If n <= 1 Then PivotDataRowsCount = 1 Else PivotDataRowsCount = n - 1
End Function

Private Sub FormatPivotValues(ByVal pt As PivotTable, ByVal fmt As String)
    Dim df As PivotField
    On Error Resume Next
    If Not pt.DataBodyRange Is Nothing Then pt.DataBodyRange.NumberFormat = fmt
    For Each df In pt.DataFields
        If InStr(1, df.Caption, "Count", vbTextCompare) > 0 Then
            df.DataRange.NumberFormat = "#,##0"
        Else
            df.DataRange.NumberFormat = fmt
        End If
    Next df
    pt.TableRange1.Font.Name = "Aptos"
    pt.TableRange1.Font.Size = 10
    On Error GoTo 0
End Sub

Private Sub SafeNumberFormat(ByVal rng As Range, ByVal fmt As String)
    On Error Resume Next
    rng.NumberFormat = fmt
    On Error GoTo 0
End Sub

Private Sub DeleteSheetIfExists(ByVal sheetName As String)
    Dim ws As Worksheet
    On Error Resume Next
    Set ws = ThisWorkbook.Worksheets(sheetName)
    On Error GoTo 0
    If Not ws Is Nothing Then ws.Delete
End Sub

Private Function LastUsedRow(ByVal ws As Worksheet) As Long
    Dim c As Range
    Set c = ws.Cells.Find(What:="", LookIn:=xlFormulas, SearchOrder:=xlByRows, SearchDirection:=xlPrevious)
    If c Is Nothing Then LastUsedRow = 1 Else LastUsedRow = c.Row
End Function

Private Function LastUsedCol(ByVal ws As Worksheet) As Long
    Dim c As Range
    Set c = ws.Cells.Find(What:="", LookIn:=xlFormulas, SearchOrder:=xlByColumns, SearchDirection:=xlPrevious)
    If c Is Nothing Then LastUsedCol = 1 Else LastUsedCol = c.Column
End Function

Private Function HeaderColumn(ByVal ws As Worksheet, ByVal headerText As String) As Long
    Dim i As Long, lastCol As Long

```

```

lastCol = LastUsedCol(ws)
For i = 1 To lastCol
    If LCase$(Trim$(CStr(ws.Cells(1, i).Value))) = LCase$(Trim$(headerText)) Then
        HeaderColumn = i
        Exit Function
    End If
Next i
HeaderColumn = 0
End Function

Private Sub RequireHeader(ByVal ws As Worksheet, ByVal headerText As String)
    If HeaderColumn(ws, headerText) = 0 Then
        Err.Raise vbObjectError + 200, , "Missing required column: " & headerText
    End If
End Sub

```